

Mathematical Logic

→ Proposition :-

proposition is a declarative statement which is either true or false but not both.

Propositions are usually denoted by small letters such as p, q, r, s, t , etc.

- If the proposition is true then it is denoted by capital T or 1 .
- If the proposition is false then it is denoted by capital F or 0 .

Example :- $q = \sqrt{5}$ is an integer $\Rightarrow F$ (False)

$p = 2 + 3 = 6 \Rightarrow F$ (False)

$r = 3 + 3 = 6 \Rightarrow T$ (True)

→ Logical connectives :-

The words like and, or, not, if then, if and only if make are called logical connectives.

Any statement contains these logical connectives is called compound proposition.

The proposition which is free from the logical connectives is called simple proposition.

Example :- $p = \text{Roses are red}$

$q = \text{violetes are blue}$

→ roses are red and violetes are blue.

→ Negation :- $[\sim]$

If p is a proposition then negation of p is not p and it is denoted by negation p ($\sim p$) or $(\neg p)$.

If the truth value of p is true then the value of not p ($\neg p$) is false and vice-versa.

p	$\neg p$
T	F
F	T

→ AND [conjunction] (" $\wedge$ ")

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0

Let p and q be the simple propositions then conjunction of p and q is ~~also~~ denoted by $p \wedge q$.
The truth value of $p \wedge q$ is true if both p and q are true otherwise it is false.

→ Disjunction [OR] (" $\vee$ ")

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

$$1 + 1 = 1$$

$$1 + 0 = 1$$

$$0 + 1 = 1$$

$$0 + 0 = 0$$

Let ~~the~~ p and q be simple propositions then disjunction of p and q is ' $p \vee q$ ' and it is denoted by (" $p \vee q$ ").
The truth value of $p \vee q$ is true if any one of the p and q is true otherwise if both are false the $p \vee q$ is false.

→ Construct the truth table for following compound propositions

(i) $p \vee \neg q$

p	q	$\neg q$	$p \vee \neg q$
T	T	F	T
T	F	T	T
F	T	F	F
F	F	T	T

(ii) $p \wedge \neg q$

p	q	$\neg q$	$p \wedge \neg q$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

(iii) $(\neg p \wedge \neg q) \vee (\neg p \vee q)$

p	q	$\neg p$	$\neg q$	$(\neg p \wedge \neg q)$	$(\neg p \vee q)$	$(\neg p \wedge \neg q) \vee (\neg p \vee q)$
T	T	F	F	F	T	T
T	F	F	T	F	F	F
F	T	T	F	F	T	T
F	F	T	T	T	T	T

(iv) $(p \wedge q) \vee (p \wedge \neg q)$

p	q	$p \wedge q$	$p \wedge \neg q$	$(p \wedge q) \vee (p \wedge \neg q)$
T	T	T	F	T
T	F	F	T	T
F	T	F	F	F
F	F	F	F	F

(v) $(p \vee q) \wedge (p \vee q)$

p	q	$(p \vee q)$	$(p \vee q)$	$(p \vee q) \wedge (p \vee q)$
T	T	T	T	T
T	F	T	T	T
F	T	T	T	T
F	F	F	F	F

(vi) $\neg p \vee \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

(vii) $\neg p \wedge \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

→ Conditional [if then] :- ' $\rightarrow$ '
 Let p and q be simple propositions then conditional of p and q is 'if p then q' and it is denoted by ' $p \rightarrow q$ '.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

The truth table of ' $p \rightarrow q$ ' is false if p is true and q is false otherwise it is true.

→ bi-conditional [iff] :- $[\leftrightarrow]$

Let p and q be simple propositions, then bi-conditional of p and q is $p \leftrightarrow q$ and it is denoted by $p \leftrightarrow q$.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

The truth value of ' $p \leftrightarrow q$ ' is true only when both p and q are true or false.

→ Tautology (T_0)

A compound proposition which is true for all possible truth values of its components is called Tautology.

$p \vee \sim p$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

→ Contradiction (F_0)

The compound proposition which is false for all possible truth value of its components is called contradiction.

Example :- $p \wedge \sim p$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

→ Contingency :-

A compound proposition that can be true or false that is neither tautology nor contradiction depending upon the truth values of its components is called contingency.

p v q		
p	q	p v q
T	T	T
T	F	T
F	T	T
F	F	F

→ Logical Equivalence

Let 'u' and 'v' be two compound propositions which are said to be logically equivalent whenever u and v have the same truth value. It is denoted by " $\iff$ ".

Ex:- $\neg(p \vee q) \iff \neg p \wedge \neg q$

			(u)				(v)
p	q	p v q	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	
T	T	T	F	F	F	F	
T	F	T	F	F	T	F	
F	T	T	F	T	F	F	
F	F	F	T	T	T	T	

hence $u \iff v$

$\neg(p \vee q) \iff \neg p \wedge \neg q$

Construct the truth table for the following

1) $[(p \wedge q) \vee \sim H] \leftrightarrow p$

p	q	H	$(p \wedge q)$	$\sim H$	$(p \wedge q) \vee \sim H$	$\leftrightarrow p$
T	T	T	T	F	T	T
T	T	F	T	T	T	T
T	F	T	F	F	F	F
T	F	F	F	T	T	T
F	T	T	F	F	F	T
F	T	F	F	T	T	F
F	F	T	F	F	F	T
F	F	F	F	T	T	F

2) Prove that $[(\sim q) \wedge (p \rightarrow q)] \rightarrow \sim p$ is a (T_0)

p	q	$\sim q$	$(p \rightarrow q)$	$(\sim q) \wedge (p \rightarrow q)$	$\sim p$	$a \rightarrow \sim p$
T	T	F	F	F	F	T
T	F	T	T	T	F	T
F	T	F	T	F	T	T
F	F	T	F	F	T	T

3) Show that for any proposition p and q, the compound proposition $p \leftrightarrow (p \vee q)$ is a tautology and the compound proposition $p \wedge (\sim p \wedge q)$ is a contradiction.

p	q	$p \vee q$	$p \rightarrow (p \vee q)$	$\sim p$	$\sim p \wedge q$	$p \wedge (\sim p \wedge q)$
T	T	T	T	F	F	F
T	F	T	T	F	F	F
F	T	T	T	T	T	F
F	F	F	T	T	F	F

$$p \rightarrow (p \vee q) = T_0 \quad , \quad p \wedge (\sim p \wedge q) = F_0$$

4) Prove that $[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$ is a T₀

p	q	r	(p → q) ^(a)	(q → r) ^(b)	(p → q) ∧ (q → r) ^(a)	p → r ^(b)	a → b
T	T	T	T	T	T	T	T
T	T	F	T	F	F	T	T
T	F	T	F	T	F	F	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

5) Prove that $\{p \rightarrow (q \rightarrow r)\} \rightarrow \{(p \rightarrow q) \rightarrow (p \rightarrow r)\}$ is a T₀

p	q	r	q → r ^(a)	p → (q → r) ^(a)	p → q ^(b)	p → r ^(b)	a → b
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	T
T	F	F	T	T	F	F	T
T	F	T	T	T	F	T	T
F	F	T	T	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	F	T	T	T	T	T

6) Show that $\underbrace{\sim(p \vee q)}_{(u)} \iff \underbrace{\sim p \wedge \sim q}_{(v)}$

p	q	~p ^(u)	~q ^(u)	(p ∨ q) ^(u)	~(p ∨ q) ^(u)	~p ∧ ~q ^(v)
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

∴ It is logically equivalent.

7) Let p and q be any 2 statements for which the conditional $p \rightarrow q$ is False. Determine the truth values of the following compound proposition.

- (i) $p \vee q$
- (ii) $\sim p \vee q$
- (iii) $q \rightarrow p$
- (iv) $\sim q \rightarrow \sim p$

$$\therefore p = T$$

$$q = F$$

$p \rightarrow q = F$
By using the definition of conditional

$$\begin{aligned} \text{(i)} \quad p \vee q \\ \Rightarrow T \vee F \\ \Rightarrow T \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \sim p \vee q \\ \Rightarrow \sim(T) \vee F \\ \Rightarrow F \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad q \rightarrow p \\ \Rightarrow F \rightarrow T \\ \Rightarrow T \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \sim q \rightarrow \sim p \\ \Rightarrow \sim(F) \rightarrow \sim(T) \\ \Rightarrow T \rightarrow F \\ \Rightarrow F \end{aligned}$$

8) Given p is true and q is false find the truth value.

$$\text{(i)} \quad \sim(p \wedge q) \vee \sim(q \rightarrow p)$$

$$\text{(ii)} \quad (p \rightarrow q) \vee \sim(p \leftrightarrow \sim q)$$

$$p = T, q = F$$

$$\begin{aligned} \text{(i)} \quad \sim(p \wedge q) \vee \sim(q \rightarrow p) \\ \Rightarrow \sim(F) \vee \sim(T) \\ \Rightarrow T \vee F \\ \Rightarrow T \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad (p \rightarrow q) \vee \sim(p \leftrightarrow \sim q) \\ \Rightarrow (T \rightarrow F) \vee \sim(T \leftrightarrow \sim(F)) \\ \Rightarrow F \vee F \\ \Rightarrow F \end{aligned}$$

- 9) Indicate how many rows are needed in the truth table for the compound proposition $(p \vee q) \leftrightarrow [(r \wedge s) \rightarrow t]$. Find the truth value of proposition if p and r is true and q, s, t are false.

$$p = T$$

$$r = T$$

$$q = F$$

$$s = F$$

$$t = F$$

$$(T \vee F) \leftrightarrow [(T \wedge F) \rightarrow F]$$

$$\Rightarrow (T \vee T) \leftrightarrow [(F \wedge F) \rightarrow F]$$

$$\Rightarrow T \leftrightarrow F \rightarrow F$$

$$\Rightarrow T \leftrightarrow T$$

$$\Rightarrow T$$

- * Given compound proposition contains $n = 5$ propositions therefore the number of rows required in the truth table is $2^n = 2^5 = 32$ rows.

- 10) Prove that $[(p \vee q) \rightarrow r]$ is logically equivalent to $[(p \rightarrow r) \wedge (q \rightarrow r)]$

(u)					(v)		
p	q	r	$(p \vee q)$	$(p \vee q) \rightarrow r$	$p \rightarrow r$	$q \rightarrow r$	$[(p \rightarrow r) \wedge (q \rightarrow r)]$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	T	T	F	T	T
T	F	F	T	F	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

$$\therefore (p \vee q) \rightarrow r \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)]$$

$u \Leftrightarrow v$

11) Prove that $[(p \vee q) \rightarrow r] \leftrightarrow [\neg r \rightarrow \neg(p \vee q)]$ is T_0 .

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \rightarrow r$	$\neg(p \vee q)$	$\neg r \rightarrow \neg(p \vee q)$	$u \leftrightarrow v$
T	T	T	T	F	T	F	T	T
T	T	F	T	T	F	F	F	T
T	F	T	T	F	T	F	T	T
T	F	F	T	T	F	F	F	T
F	T	T	T	F	T	F	T	T
F	T	F	T	T	F	F	F	T
F	F	T	F	F	T	T	T	T
F	F	F	F	T	T	T	T	T

$\therefore u \leftrightarrow v = T_0$

Note:-

- 1) Converse $(q \rightarrow p)$
- 2) Inverse $(\neg p \rightarrow \neg q)$
- 3) Contrapositive $\neg q \rightarrow \neg p$

Q:- Write converse, inverse and contrapositive of conditional

- 1) p : Quadrilateral is a parallelogram
- 2) q : diagonals of Quadrilateral bisect each other.

sol:- In conditional

$(p \rightarrow q)$: if Quadrilateral is a parallelogram then diagonals of quadrilateral bisect each other.

Converse: $(q \rightarrow p)$

If diagonals of Quadrilateral bisect each other then Quadrilateral is a parallelogram.

Inverse: $(\neg p \rightarrow \neg q)$

If Quadrilateral is not a parallelogram then the diagonals of Quadrilateral does not bisect each other.

Contrapositive: $(\sim q \rightarrow \sim p)$

If diagonals of quadrilateral does not bisect each other then quadrilateral is not a parallelogram.

Q:- p : Triangle is not isosceles
 q : Triangle is not equilateral

Conditional

$(p \rightarrow q)$: If triangle is not isosceles then triangle is not equilateral.

Converse: $(q \rightarrow p)$

If triangle is not equilateral then triangle is not isosceles.

Inverse: $(\sim p \rightarrow \sim q)$

If triangle is isosceles then triangle is equilateral.

Contrapositive: $(\sim q \rightarrow \sim p)$

If triangle is equilateral then triangle is isosceles.

→ open statement

A statement which becomes a proposition when the values of the variable involved is known
open statement.

Example :- $p(x): x + 3 = 5$ (open statement)
 $x = 0$

$$p(0): 0 + 3 = 5 = F$$

$$x = 1$$

$$p(1)$$

$$p(1): 1 + 3 = 5 = F$$

$$x = 2$$

$$p(2): 2 + 3 = 5 = T$$

Proposition.

→ Quantified statement
A proposition involving the idea of quantity is known as quantified statement such as: for all, for every, for some, there exist etc.

there are 2 types of quantifiers:-

(i) Universal Quantifier.

The words like for all, for every, for each are involved in the sentence is called universal quantifiers. It is denoted by $(\forall)$

(ii) Existential Quantifier.

The words like for some, there exist, at least are involved in the sentence is called existential quantifiers. It is denoted by $(\exists)$

note:-

i) $[\forall x, p(x)]$ is true only when $p(x)$ is True for every $x \in U$.

(ii) $[\exists x, p(x)]$ is false only when $p(x)$ is False for every $x \in U$.

(iii) Negation of the Quantified statement

$$\sim[\forall x, p(x)] \equiv \exists x [\sim p(x)]$$

$$(iv) \sim[\exists x, p(x)] \equiv \forall x, [\sim p(x)]$$

$$\begin{bmatrix} \sim \forall = \exists \\ \sim \exists = \forall \end{bmatrix}$$

ϕ :- let the universal set be the set of all integers
consider the following open statement ϕ

$$p(x): x \leq 3$$

$$q(x): x+1 \text{ is odd}$$

$$r(x): x \leq 0$$

find the truth value of the following.

(i) $p(2)$
sol:- consider $p(x): x \leq 3$

$$\text{put } x = 2$$

$$p(2): 2 \leq 3 \Rightarrow \text{True.}$$

$$(i) p(3) \vee \neg q(3)$$

$$p(x): x \leq 3$$

$$(put\ x=3) p(3): 3 \leq 3 \Rightarrow \text{True (T)}$$

$$q(x): x \leq 0$$

$$put\ x=3$$

$$q(3): 3 \leq 0 \Rightarrow \text{False (F)}$$

$$\neg q(3): T$$

$$\Rightarrow T \vee T \Rightarrow T$$

$$(iii) \neg p(4) \wedge \neg q(5)$$

$$p(x): x \leq 3$$

$$put\ x=4$$

$$p(4): 4 \leq 3 \Rightarrow \text{False (F)}$$

$$\neg p(4) \Rightarrow \text{True (T)}$$

$$q(5): 5 + 1 \text{ is odd} = F$$

$$\neg q(5): T$$

$$\Rightarrow T \wedge T = T$$

$$(iv) p(-1) \wedge q(1)$$

$$p(x): x \leq 3$$

$$put\ x = -1$$

$$p(-1): -1 \leq 3 \Rightarrow T$$

$$q(x): x \leq 3$$

$$put\ x = 1$$

$$q(1): 1 \leq 3 \Rightarrow F$$

$$\Rightarrow T \wedge F = F$$

$$(v) p(0) \rightarrow q(0)$$

$$\Rightarrow T \rightarrow T$$

$$\Rightarrow T$$

Q: For the set of all integers $p(x)$ is $x > 0$ (positive no),
 $q(x)$ is x is even

$r(x)$: x is a perfect square

$s(x)$: x is divisible by 3.

$t(x)$: x is divisible by 7

using the above propositions write down the following quantified statements in symbolic form.

(i) at least 1 integer is even

$$\rightarrow \exists x, q(x)$$

(ii) There exist a positive integer and that is even

$$\rightarrow \exists x, p(x) \wedge q(x)$$

(iii) Some even integers are divisible by 3.

$$\rightarrow \exists x, q(x) \wedge s(x)$$

(iv) If x is even and a perfect square then x is not divisible by 3.

$$\rightarrow \forall x, q(x) \wedge r(x) \rightarrow \sim s(x)$$

[if $\Rightarrow$ $\forall$]
[$\rightarrow$ then]

(v) If x is odd or x is not divisible by 7 then x is divisible by 3.

$$\rightarrow \forall x, \sim q(x) \vee \sim t(x) \rightarrow s(x)$$

Q:- Express the following symbolic statements in words and indicate the truth value.

(i) $\forall x, [r(x) \rightarrow p(x)]$, where $x = 0$.

$\rightarrow$ For all integer x , if x is a perfect square then $x > 0$

$r(0) = 0$ is a perfect square

$$p(0) = 0 > 0 \equiv F$$

(ii) $\exists x, [s(x) \wedge \neg q(x)]$ where $x=9$

$\Rightarrow$ For some integer x , x is divisible by 3 and x is odd.

$$[s(9) \wedge \neg q(9)] = T \wedge T \Rightarrow T$$

(iii) $\forall x, [\neg r(x)]$ where $x=4$

$\Rightarrow$ For all integer x , x is ^{not a} perfect square.

$$[\neg r(4)] = F$$

(iv) $\forall x, [h(x) \vee t(x)]$ where $x=3$

$\Rightarrow$ For all integer x , x is a perfect square or x is divisible by 7.

$$[h(3) \vee t(3)]$$

$$\Rightarrow F \vee F \Rightarrow F$$